PDR WORLD

MASTER PRODUCT CATALOGUE 2024–2025

Empowering Global Connectivity

Discover our comprehensive suite of state-of-the-art optical fiber solutions, carrier-grade passive and active components, advanced cable management systems, and precision testing equipment.

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- Optical Line Protection System (OLPS)

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ACTIVE COMPONENTS > DAC & AOC

Active Optical Cable (AOC)

Active Optical Cable, a reliable, low-power alternative to transceiver assemblies.

A reliable, cost-effective alternative to separate transceiver-and-fiber assemblies, available in lengths from 0.5 m to 100 m.

Key Features

- Available in lengths from 0.5 m to 100 m
- Low power consumption, maximum 1.9 W per end
- Hot-swappable QSFP28 form factor with high-strength pull-tab latch
- Flexible OM3 multimode fiber with slim 3.0 mm outer diameter

Applications

- Point-to-point connections within data centres
- High-performance computing and storage racks
- Rack-to-rack and inter-row top-of-rack architectures

Technical Specifications

Parameter / Specification	Value / Detail
Operating Wavelength	850 nm
Fiber Type	OM3 multimode
Available Legths	0.5m to 100m
Compliance	QSFP28 MSA, IEEE P802.3bm, 100GBASE-SR4, RoHS/REACH, TÜV/UL

PASSIVE COMPONENTS

Variable Fiber Attenuator

Variable optical attenuator for precise in-line signal level control.

PDR Variable Fiber Optic Attenuator (VOA) provides continuously adjustable attenuation up to 30 dB for single-mode and multimode fiber with accurate, low-loss performance across 1260–1650 nm for telecom, CATV, data center, and test applications.

Key Features

- Wide attenuation range of 0 to 30 dB with precise, continuous adjustment capability
- High attenuation accuracy of ±0.5 dB for reliable signal level management
- Low insertion loss of ≤ 1.0 dB to minimise signal degradation in the optical path
- High return loss of ≥ 50 dB (SM) for excellent back-reflection suppression
- Available in SC, LC, FC, and ST connector interfaces for universal compatibility
- Supports both single-mode and multi-mode fiber types

Applications

- Fiber-to-the-Home (FTTH) and Fiber-to-the-Building (FTTB) network power balancing
- Telecommunications network testing, commissioning, and maintenance
- Cable TV (CATV) and broadband infrastructure signal level optimisation
- Data centre interconnects and optical link budget management

Technical Specifications

Parameter / Specification	Value / Detail
Attenuation Range	0 to 30 dB (Typical)
Attenuation Accuracy	± 0.5 dB
Resolution	0.1 dB (Typical)
Insertion Loss	≤ 1.0 dB
Return Loss	≥ 50 dB (SM)
Wavelength Range	1260 to 1650 nm
Connector Type	SC / LC / FC / ST
Fiber Type	Single-mode / Multi-mode
Polarisation Dependent Loss (PDL)	≤ 0.2 dB
Repeatability	≤ ± 0.2 dB
Operating Temperature	-20°C to +70°C
Storage Temperature	40°C to +85°C
Durability	≥ 1000 adjustment cycles

PASSIVE COMPONENTS

Bare Fiber Adapter

Bare fiber adapter for quickly coupling unterminated fiber to test equipment.

PDR Bare Fiber Adapters enable tool-free temporary connection of stripped bare fibers to SC, LC, FC, ST, or SMA interfaces for reliable testing, field, and lab applications.

Key Features

- Available in SC, LC, FC, ST, and SMA interface types to suit all common test equipment
- Secure mechanical clamping mechanism holds bare fiber firmly without adhesive or tools
- Compatible with both single-mode (OS1/OS2) and multimode (OM1, OM4) optical fibers
- Compact, lightweight body for easy handling in lab and field environments
- Simple plug-and-play operation, insert bare fiber on one side, connect to equipment on the other
- Enables fast temporary termination for fiber testing and measurement applications

Applications

- Fiber-to-the-Home (FTTH) and Fiber-to-the-Building (FTTB) network testing and commissioning
- Telecommunications network measurement, fault location, and maintenance
- OTDR and optical power meter connections during cable installation and characterisation
- Data centre and enterprise LAN fiber testing and troubleshooting

Technical Specifications

Parameter / Specification	Value / Detail
Adapter Types	SC, LC, FC, ST, SMA
Compatible Fiber	Single-mode (OS1/OS2) | Multimode (OM1, OM2, OM3, OM4)
Fiber Cladding Diameter	125 µm (standard)
Fiber Holding Mechanism	Mechanical clamp, no adhesive or tools required
Insertion Loss	SC/FC/ST: ≤ 0.5 dB | LC: ≤ 0.5 dB | SMA: ≤ 1.0 dB (typical)
Return Loss	SC/FC/ST: ≥ 20 dB | LC: ≥ 20 dB | SMA: ≥ 14 dB
Mating Durability	≥ 500 mating cycles
Housing Material	Zinc alloy body with precision ceramic / stainless insert
Operating Temperature	20°C to +70°C
Storage Temperature	-40°C to +85°C
Operating Humidity	5%, 95% RH (non-condensing)
Bare Fiber Insertion	Simple straight insertion, no stripping jig required
Applications	Lab testing, field measurement, emergency restoration, PON/OTDR use
Compliance	RoHS compliant

PASSIVE COMPONENTS

Cassette Cleaner

Reel cassette cleaner for safe, lint-free connector end-face cleaning.

PDR Fiber Optic Connector Cleaner: Compact handheld cleaner for SC, FC, ST, MU, LC, MT, D4, DIN & E2000 connectors, using non-alcohol superfine fiber tape for scratch-free, residue-free cleaning with 500+ cycles per replaceable cartridge.

Key Features

- Compatible with SC, FC, ST, MU, LC, MT, D4, DIN, and E2000 fiber optic connectors
- Special non-alcohol superfine fiber tape for safe, residue-free end-face cleaning
- Anti-static design prevents re-contamination after each cleaning stroke
- Over 500 cleaning cycles per tape cartridge, low cost per clean
- Replaceable tape cartridge design minimises ongoing consumable costs
- Simple one-push action for consistent, repeatable cleaning results

Applications

- Fiber-to-the-Home (FTTH) and Fiber-to-the-Building (FTTB) connector maintenance
- Telecommunications equipment room and outside plant installation and servicing
- Data centre patch panel, switch port, and transceiver connector cleaning
- Cable TV (CATV) and broadband network end-face preparation and maintenance

Technical Specifications

Parameter / Specification	Value / Detail
Compatible Connectors	SC, FC, ST, MU, LC, MT, D4, DIN, E2000
Cleaning Medium	Non-alcohol superfine fiber tape
Cleaning Times per Cartridge	500+ cycles
Replacement Cartridge	Available (field-replaceable)
Anti-static	Yes
Cleaning Action	One-push mechanical advance
Dimensions (L×W×H)	115 mm × 79 mm × 32 mm
Weight	Lightweight, suitable for field carry
Solvents / Liquids Required	None

CABLE MANAGEMENT DEVICES

CAT 6 Patch Panel

24-port Cat 6 patch panel for organised copper network termination.

PDR Cat 6 24-Port Distribution Panel features 24 RJ-45 IDC ports with T568A/T568B support, meeting ISO/IEC 11801 and EIA/TIA standards for Gigabit and 10GBase-T structured cabling.

Key Features

- 24-port RJ-45 panel conforming to ISO/IEC 11801 and EIA/TIA 568A & 568B standards
- Metallic panel (SPCC) with fireproof ABS UL-94V0 plastic body for durability and safety
- PCB: FR-4 / 1.6 mm UL-94V0 for superior signal integrity
- Supports Cat 6 performance, suitable for Gigabit Ethernet and 10G-Base-T applications
- RJ-45 jack housing: PBT / 8P8C with 50 micro-inch gold-plated contacts for reliable connectivity
- Supports both T568A and T568B colour-coding with clearly labelled port identification

Applications

- Data centre patch rooms and server rack cable management
- Office building and campus network distribution frames
- Enterprise LAN and structured cabling infrastructure
- Telecommunications rooms and main distribution frames (MDF / IDF)

Technical Specifications

Parameter / Specification	Value / Detail
Standard Compliance	ISO/IEC 11801 | EIA/TIA 568A & 568B
Number of Ports	24 x RJ-45
Category	Cat 6
Panel Material	Metallic, SPCC (Steel, Cold-rolled, Coil)
Plastic Body	Fireproof ABS UL-94V0
RJ-45 Jack Housing	PBT / 8P8C, 50 micro-inch gold plated contacts
PCB	FR-4 / 1.6 mm UL-94V0
IDC Body	Fireproof ABS UL-94V0
IDC Pin Material	Phosphor Bronze, Tinned
IDC Wire Range	22 to 26 AWG solid conductor
Termination Tool	110 or Krone punch-down tool
Color Coding	T568A and T568B (Arrange: 8 7 6 3 2 1 4 5)
Insulation Resistance	> 500 MΩ (minimum)
DC Resistance	< 0.1 Ω (maximum)

PASSIVE COMPONENTS

CAT 6 Patch Cord

High-speed copper patch cord

The PDR U/UTP Cat 6 Patch Cord provides flexible, reliable high-speed Ethernet connectivity with Cat 6 RJ45 connectors and PVC or LSZH jacket options.

Key Features

- U/UTP 4-pair construction with stranded 24 AWG (7×0.18 mm) conductors for flexibility and durability
- Cat 6 RJ45 plug termination with moulded PVC strain relief boot, colour-matched to outer jacket
- Jacket outer diameter 5.8 ± 0.20 mm with ripcord for easy stripping during termination
- HDPE insulation for low capacitance and consistent electrical performance across all four pairs
- Outer jacket available in PVC or LSZH, specify at time of order to suit installation requirements
- Pair colour coding: Blue/White, Orange/White, Green/White, Brown/White

Applications

- 100 Base-TX and 100 Base-T (100VG-AnyLAN) Fast Ethernet network connections
- 1000 Base-TX high-speed enterprise LAN patch panel deployments
- 1000 Base-T Gigabit Ethernet structured cabling installations
- Data centre patch panel and server rack interconnections requiring flexible, durable patch cords

Technical Specifications

Parameter / Specification	Value / Detail
Construction	U/UTP (Unshielded Twisted Pair)
Pair Count	4 pairs
Conductor	Stranded 24 AWG (7×0.18 mm)
Insulation Material	HDPE
Insulation Diameter	0.91 ± 0.01 mm
Pair Colours	Blue/White, Orange/White, Green/White, Brown/White
Cross Filler	Yes
Connector	Cat 6 RJ45 Plug
Strain Relief Boot	Moulded PVC, colour-matched to jacket
Outer Jacket Material	PVC or LSZH (specify at order)
Jacket Thickness	0.6 ± 0.05 mm
Outer Diameter	5.8 ± 0.20 mm
Ripcord	Yes
Rated Temperature	60°C, 75°C, 90°C

PASSIVE COMPONENTS

Fiber Optic Cleaner Pen

One-click cleaner pen for fast, reliable connector end-face cleaning.

PDR Fiber Optic Cleaner Pen is a compact, single-action cleaner for female connectors, removing dust, oil, and contaminants without scratching, with anti-static microfiber technology for 800+ cleanings and optimal optical performance.

Key Features

- Push-to-clean operation, easy single-motion engagement initiates the cleaning cycle
- 800+ cleanings per unit, cost-effective and long lasting
- Anti-static resin body, prevents electrostatic charge build-up during cleaning
- Densely stranded, debris-free micro-fiber cleaning element for thorough end-face cleaning
- Extendable tip design reaches recessed connectors on panels and equipment
- Cleaning system rotates 180° for a full, complete sweep of the ferrule end face

Applications

- Fiber network panels and patch panel assemblies in data centres
- Outdoor FTTX (Fiber-to-the-X) network installations and field maintenance
- Cable assembly production facilities requiring consistent end-face cleanliness
- Testing laboratories and quality-control environments for fiber optic components

Technical Specifications

Parameter / Specification	Value / Detail
Compatible Connectors	2.5 mm: FC, SC, ST | 1.25 mm: LC, MU | 2.0 mm: SMPTE
Ferrule Tip Diameter	2.5 mm (CLEP 2.5) | 1.25 mm (CLEP 1.25) | 2.0 mm (CLEP 2.0)
Cleaning Method	Push-to-clean rotary mechanism
Cleaning Rotation	180° full sweep per actuation
Cleanings Per Unit	800+ cleanings
Cleaning Element	Densely stranded micro-fiber, debris-free
Tip Design	Extendable, reaches recessed connectors
Engagement Indicator	Audible click on correct engagement
Body Material	Anti-static resin
Operation	Single push motion, no batteries required
Disposal	Disposable, replace when cleanings are exhausted

PASSIVE COMPONENTS

CPRI Patchcord

Outdoor waterproof fiber optic assembly

PDR CPRI Outdoor Waterproof Patch Cords feature rugged IP-rated, blind-mate fiber assemblies with multiple waterproof connector options, armored/non-armored cable designs, and 2–24 core single-mode or multimode fiber for harsh outdoor applications.

Key Features

- Waterproof connector housings: ODVA, IP, FX, PDLC, and NSN types with MPO/MTP, LC, or SC ferrules
- IP-rated blind-mating design for quick, tool-free installation in field and outdoor environments
- Robust bayonet locking mechanism for easy, fast, and secure connector mating and retention
- Lightweight, waterproof, dustproof, and corrosion-resistant construction for harsh environments
- Available in flat drop or round cable, armored or non-armored, for flexible routing requirements
- Single mode fiber: G.652D, G.657A1, G.657A2, G.657B3, multimode: OM1, OM2, OM3, OM4

Applications

- WiMAX and LTE base station fiber backhaul and fronthaul cabling in outdoor environments
- Remote Radio Head (RRH) and Remote Radio Unit (RRU) interconnections in distributed antenna systems
- Industrial outdoor installations in factories, plants, and process control environments
- Harsh environment deployments requiring waterproof, dustproof, and corrosion-resistant terminations

Technical Specifications

Parameter / Specification	Value / Detail
Cable Diameter	5.0 mm | 7.0 mm | 8.0 mm (±0.3 mm)
Connector Housing Types	ODVA | IP | FX | PDLC | NSN
Ferrule / Connector Types	MPO/MTP | LC | SC
Fiber Mode, Single-Mode	G.652D | G.657A1 | G.657A2 | G.657B3
Fiber Mode, Multimode	OM1 | OM2 | OM3 | OM4
Fiber Count	2 / 4 / 8 / 12 / 24 core
Cable Jacket	LSZH | TPU | PE
Cable Length	1 to 150 m; custom lengths available
Operating Temperature	-40°C to +85°C
SM Insertion Loss, Typical	≤0.30 dB (IEC 61300-3-34)
SM Insertion Loss, Maximum	≤0.75 dB (IEC 61300-3-34)
MM Insertion Loss, Typical	≤0.50 dB (IEC 61300-3-34)
MM Insertion Loss, Maximum	≤0.75 dB (IEC 61300-3-34)
SM Return Loss	≥50 dB (PC) | ≥60 dB (APC) (IEC 61300-3-6)

PASSIVE COMPONENTS

CWDM Mux / DeMux Module

CWDM multiplexer / demultiplexer modules for cost-effective wavelength expansion.

PDR CWDM Mux/DeMux offers 4/8/16-channel CWDM (1270–1610 nm) in ABS, LGX, and 1U rack formats with low loss, high isolation, and reliable optical performance.

Key Features

- Four form factors: ABS Compact Module (two sizes), 1U Rack Equipment, and LGX Module
- Supports 4, 8, and 16 channel configurations across the CWDM grid from 1270 nm to 1610 nm
- Low insertion loss: ≤1.6 dB (4ch), ≤2.8 dB (8ch), ≤4.5 dB (16ch) for ABS Compact Module
- High adjacent channel isolation >30 dB; non-adjacent channel isolation >40 dB
- Low passband ripple ≤0.40 dB and polarization dependent loss <0.30 dB across all variants
- Return loss >45 dB and directivity >50 dB for excellent signal integrity

Applications

- CATV and cable television headend signal multiplexing and distribution across CWDM wavelength bands
- Metro and access WDM network capacity expansion using the full 1270 to 1610 nm CWDM grid
- Optical network monitoring and per-wavelength channel management in carrier-grade systems
- Test and measurement equipment for wavelength-selective optical characterisation

Technical Specifications

Parameter / Specification	Value / Detail
CWDM Wavelength Grid	1270, 1290, 1310 ... 1570, 1590, 1610 nm (20 nm channel spacing)
Channel Configurations	4 Channel | 8 Channel | 16 Channel
IL, ABS Compact Module (Narrow PB)	4ch: ≤1.6 dB | 8ch: ≤2.8 dB | 16ch: ≤4.5 dB
IL, ABS Compact Module (Wide PB)	4ch: ≤2.0 dB | 8ch: ≤3.6 dB | 16ch: ≤5.0 dB
IL, 1U Rack Equipment	4ch: ≤1.8 dB | 8ch: ≤3.0 dB | 16ch: ≤4.8 dB
IL, LGX Module	4ch: ≤1.8 dB | 8ch: ≤3.0 dB | 16ch: ≤4.8 dB
Passband, ABS Compact (Narrow PB)	±0.11 nm (@−0.5 dB)
Passband, ABS Compact (Wide PB)	±6.5 nm (@−0.5 dB)
Passband, 1U Rack / LGX Module	±7.5 nm (@−0.5 dB)
Isolation, Adjacent Channel	> 30 dB
Isolation, Non-Adjacent Channel	> 40 dB
Passband Ripple	≤0.40 dB
Polarization Dependent Loss (PDL)	< 0.30 dB

Parameter / Specification	Value / Detail
Polarization Mode Dispersion (PMD)	< 0.10 ps

ACTIVE COMPONENTS > DAC & AOC

Direct Attached Cable (DAC)

Passive SFP+ twinax copper cable for short-reach 10G interconnects, zero power, zero heat.

PDR SFP+ Direct Attach Cables are passive, hot-swappable, MSA-compliant 10 Gbps twinax copper cables for reliable short-reach connectivity, factory-tested and RoHS/REACH compliant.

Key Features

- Passive twinax copper, zero power, no heat generation
- Supports 10 Gbps (10GBASE-CR, 10G FCoE, 10G InfiniBand)
- Broad compatibility with switches, servers and storage
- Available in 1 m, 2 m, 3 m and 5 m lengths
- Hot-swappable SFP+ connectors, SFP+ MSA compliant
- 100% factory-tested before shipment

Applications

- Top-of-rack and rack-to-rack switching
- Enterprise server-to-switch links
- Data-centre and HPC short-reach interconnects

Technical Specifications

Parameter / Specification	Value / Detail
Cable Type	Passive twinax copper
Connector Type	SFP+ (both ends)
Nominal Data Rate	10.3125 Gbps
Available Lengths	1 m | 2 m | 3 m | 5 m
Link Distance	Up to 7 m (passive)
Minimum Bend Radius	30 mm
Operating Temperature	0°C to 70°C
Power Consumption	Passive, zero power draw
Compliance	SFP+ MSA, IEEE 802.3ae, 10GFC, RoHS/REACH, TÜV/UL

SPECIALTY DRONES

Ground unit

Ground-side fiber optical terminal that outputs video and links to the remote controller, with selectable CRSF/SBUS protocols.

Ground-side receive terminal that recovers the 1.25 Gbps single-fiber link and outputs composite video to a display while interfacing with the remote controller, image-transmission module or FPV receiver, with dial-code selectable GH1.25 or cable modes and CRSF/SBUS protocol support.

Key Features

- Ground-side (Type B) receive terminal for tethered / FPV drone fiber links
- Composite video output via BNC to display or FPV receiver
- Selectable CRSF or SBUS protocol for popular FPV / racing-drone flight controllers
- Compact 105 × 70 × 28 mm housing
- Single-fiber optical reception at 1.25 Gbps over an FC interface
- Dial-code selectable GH1.25 (W) or cable (C) interface mode
- FC optical, RJ45 100M Ethernet and BNC video front-panel interfaces

Applications

- Defence ISR & secure communications
- Perimeter / border surveillance
- FPV reconnaissance & tactical missions
- EW / jam-resistant operations

Technical Specifications

Parameter / Specification	Value / Detail
Unit Type	Type B — Ground Unit
Optical Transmission Rate	1.25 Gbps
Optical Connector	FC (single fiber)
Video Channel	Composite video output (BNC)
Serial / Telemetry	TTL serial, PPM input; CRSF / SBUS support
Interface Modes	GH1.25 (W) or Cable (C), dial-code selectable
Power Input	10–26 V DC (DC-IN); GH1.25 outputs 5 V

PASSIVE COMPONENTS

DWDM Mux/DeMux Module

DWDM multiplexer / demultiplexer modules for high-capacity dense wavelength transport.

PDR DWDM Mux/DeMux supports 4/8/16-channel ITU-T C20–C60 DWDM (1520–1570 nm, 100 GHz) in ABS or 1U rack formats with low loss, >30 dB isolation, and >45 dB return loss.

Key Features

- Available in compact ABS module (100×80×10 mm / 120×80×18 mm) and 1U rack equipment form factors
- Supports 4, 8, and 16 channel configurations covering ITU-T channels C20 to C60
- Low insertion loss: ≤2.0 dB (4ch), ≤3.0 dB (8ch), ≤4.8 dB (16ch) for ABS module
- Operating wavelength range 1520 to 1570 nm with 100 GHz minimum channel spacing
- High channel isolation: >30 dB adjacent channel, >40 dB non-adjacent channel
- Low passband ripple (≤0.30 dB) and polarization dependent loss (≤0.30 dB)

Applications

- CATV and cable television headend signal multiplexing and distribution
- Dense WDM (DWDM) transport systems for long-haul and metro network capacity expansion
- Optical network monitoring and wavelength management in carrier-grade systems
- Test and measurement equipment for wavelength-selective optical characterisation

Technical Specifications

Parameter / Specification	Value / Detail
Channel Configurations	4 Channel | 8 Channel | 16 Channel
Insertion Loss, ABS Module	4ch: ≤2.0 dB | 8ch: ≤3.0 dB | 16ch: ≤4.8 dB
Insertion Loss, 1U Rack Equipment	4ch: ≤2.2 dB | 8ch: ≤3.2 dB | 16ch: ≤5.0 dB
ITU Channels	C20, C60 (C20, C21, C22 ... C58, C59, C60)
Operating Wavelength	1520 to 1570 nm
Min. Channel Spacing	100 GHz
Passband (−0.5 dB bandwidth)	±0.11 nm
Isolation, Adjacent Channel	> 30 dB
Isolation, Non-Adjacent Channel	> 40 dB
Passband Ripple	≤0.30 dB
Polarization Dependent Loss (PDL)	≤0.30 dB
Polarization Mode Dispersion (PMD)	< 0.10 ps
Return Loss	> 45 dB
Directivity	> 50 dB

TEST AND MEASUREMENT EQUIPMENT

EasyGet WiFi Wireless Fiber Endface Microscope

Wireless transmission to smartphone/tablet

Wireless transmission to smartphone/tablet

Key Features

- Wireless transmission to smartphone/tablet
- High-resolution endface imaging
- Eliminates direct eye exposure to lasers

Applications

- Connector Endface Inspection
- Patch Panel Cleaning Verification
- Transceiver Port Checking

Technical Specifications

Parameter / Specification	Value / Detail
Connectivity	WiFi / USB
Magnification	400X
App Support	iOS / Android / Windows

PASSIVE COMPONENTS

FanOut Patch Cords

Small form factor high-density assembly

PDR MTP/MPO Fanout Patch Cord enables high-density MPO/MTP-to-single-fiber breakout for up to 100G data center applications, available in multiple cable, connector, and fanout configurations.

Key Features

- Small form factor MPO/MTP breakout to LC, SC, ST, or FC pigtails in 0.9 mm, 2.0 mm simplex, duplex, or Uniboot formats
- Supports single-mode (OS1, OS2) and multimode (OM1, OM2, OM3, OM4) fiber types
- Both one-end and both-end fanout configurations for equipment-to-panel and panel-to-panel applications
- Available in 8F, 12F, 24F, 48F, 72F, 96F, and 144F fiber counts; fully custom solutions on request
- MTP/MPO connectors available in Standard and Elite (low-loss) versions for flexible power budget management
- Micro cable construction in single and double jacket options for compact, high-density routing

Applications

- Data center interconnections between active equipment, patch panels, and fiber distribution systems
- Telecommunications and datacom network backbone and horizontal cabling
- High-density 40G and 100G structured cabling architectures requiring MPO-to-LC/SC breakouts
- Optical fiber access networks including FTTx and PON deployments

Technical Specifications

Parameter / Specification	Value / Detail
MPO Insertion Loss, Typical	SM Standard: 0.35 dB | SM Low Loss: 0.20 dB MM Standard: 0.35 dB | MM Low Loss: 0.20 dB
MPO Insertion Loss, Maximum	SM Standard: 0.75 dB | SM Low Loss: 0.35 dB MM Standard: 0.60 dB | MM Low Loss: 0.35 dB
LC / SC / ST / FC Insertion Loss, Typical	SM Standard: 0.3 dB | SM Low Loss: 0.2 dB MM Standard: 0.3 dB | MM Low Loss: 0.2 dB
Return Loss	≥60 dB (APC) | ≥20 dB (PC) | ≥50 dB (UPC)
Durability	< 0.2 dB change after 200 insertions
Operating Temperature	−40°C to +75°C
Fiber Mode, Single-Mode	OS1, OS2
Fiber Mode, Multimode	OM1, OM2, OM3, OM4
MPO Connector Type	Male or Female; PC or APC end face
Single Ferrule Connector Types	LC, SC, ST, FC; PC or APC end face
Available Ferrule Counts	4F, 8F, 12F, 24F
Fiber Count Options	8F, 12F, 24F, 48F, 72F, 96F, 144F; custom available
Cable Jacket Diameter	0.9 mm | 2.0 mm | 3.0 mm | 4.5 mm | 7.0 mm | 8.0 mm
Cable Construction	Micro single jacket | Micro double jacket

CABLE MANAGEMENT DEVICES

Fiber Distribution Box (FDB)

Fiber distribution box for splitting and distributing fiber in access networks.

PDR-FDB24B Fiber Distribution Box is a compact, rugged FTTx termination enclosure supporting straight-joint, branch, and pass-through connectivity with up to 24 SC adaptors, two 1×8 LGX splitters, and separate cable routing for easy maintenance and fault identification.

Key Features

- Accommodates 1×4, 1×8, 1×16, 2×4, 2×8, and 2×16 PLC steel tube or LGX splitter modules
- IP65 protection class, suitable for both indoor and outdoor environments
- Supports wall mounting and pole mounting for flexible field deployment
- 24 drop-cable exit ports with dedicated separation of drop cables and fiber loose tubes
- Individual link testing possible without disturbing other subscriber connections
- Safe-shaped lock with front-access opening angle exceeding 120 degrees for easy maintenance

Applications

- FTTx (FTTH / FTTB / FTTC) last-mile access network splitter distribution points
- Telecom carrier and broadband service provider outdoor cabinet installations
- Pole-mounted and wall-mounted fiber distribution in urban and rural deployments
- Multi-dwelling unit (MDU) fiber termination and subscriber connectivity hubs

Technical Specifications

Parameter / Specification	Value / Detail
Model	PDR-FDB24B
Dimensions (L x W x D)	320 × 275 × 105 mm
Weight	1.36 kg
Material	High Quality Engineering Plastic
Protection Class	IP65
Mounting Options	Wall Mount / Pole Mount
Opening Angle	> 120 degrees
Cable Diameters Supported	4 - 7 mm drop cable
Total Optical Splice Capacity	24
Maximum Cable Entries	2
Mid-Span Port	1 feeder cable entry (2 ports), 1 entrance port + 1 reserved
Drop Cable Exit Ports	24 ports
Branching & Distribution	2 in, 24 out
LGX Splitter Modules	2 x (1×8)

PASSIVE COMPONENTS

G652D Fiber Spool / Fiber Drum

Precision-wound G.652D single-mode fiber spool for long-haul, FTTH and data infrastructure.

Precision-wound ITU-T G.652D single-mode reels with Germania-doped silica core, ultra-low attenuation under 0.200 dB/km at 1550 nm across 1310–1625 nm, individually tested and barcoded for traceability.

Key Features

- ITU-T G.652D compliant, meets and exceeds international single-mode fiber standards
- Ultra-low attenuation under 0.200 dB/km at 1550 nm for long-haul signal integrity
- Wide operating wavelength range, optimized from 1310 nm to 1625 nm
- Dual-layer UV-curable resin coating for mechanical and environmental protection
- Germania-doped silica core with pure silica cladding
- Precision-wound reels, individually tested and labelled with unique ID and barcode

Applications

- Long-haul telecommunications backbone
- FTTH and access network deployment
- High-bandwidth data-centre infrastructure
- Spool stock for splicing, testing and field restoration

Technical Specifications

Parameter / Specification	Value / Detail
Fiber Type	Single Mode, ITU-T G.652D
Attenuation	≤ 0.36 dB/km (1310 nm); ≤ 0.23 dB/km (1550 nm)
Cable Cutoff Wavelength	under 1260 nm
Mode Field Diameter	9.2 ±0.4 µm at 1310 nm
Cladding Diameter	125 ±0.7 µm

PASSIVE COMPONENTS

Fiber Optic Connector, Field Installable

Field-installable fast connectors for quick, tool-light on-site termination.

Field-installable SC connector for tool-light on-site termination with no adhesives or polishing, fully zirconia-ceramic ferrule, ≤0.3 dB IL, ≥45 dB (PC)/≥55 dB (APC) RL, re-assemblable.

Key Features

- Field installable without adhesives, polishing, or electrical power, ready for immediate on-site use
- Fully zirconia-ceramic ferrule: nominal 2.5 mm diameter, ±0.5 mm tolerance, ≤1 µm fiber concentricity
- Ferrule end-face geometry compliant with Telcordia GR-326 interferometric measurement standard
- Low insertion loss ≤0.3 dB and high return loss ≥45 dB (PC) / ≥55 dB (APC)
- Re-assemblable design reduces connection defect rate and simplifies field maintenance and rework
- Type-A: compact 52.4 × 9 × 7.3 mm body, compatible with 3.1 × 2 mm flat cable

Applications

- FTTx last-mile fiber drops and subscriber premises installations requiring rapid field termination
- Telecommunications access network buildouts where on-site polishing equipment is impractical
- Enterprise and campus network fiber installations demanding fast, reliable connector assembly
- Field restoration and emergency repair of optical fiber links with minimal downtime

Technical Specifications

Parameter / Specification	Value / Detail
Connector Interface	SC
Polish Type	PC / APC
Body Type	Type-A / Type-B
Ferrule Material	Fully Zirconia-Ceramic
Insertion Loss	≤0.3 dB
Return Loss	≥45 dB (PC) / ≥55 dB (APC)
Compatible Cable	3.1×2 mm flat / 3 mm drop / 2.0 mm / 0.9 mm

PASSIVE COMPONENTS

Fiber Optic Patch Cords & Pigtails

Broadest choice of patchcords in the industry

PDR Fiber Optic Patch Cords and Pigtails offer IEC/EIA/TIA/GR-326 compliant, factory-terminated singlemode and multimode assemblies with multiple connector, cable, and fiber options.

Key Features

- Widest connector selection: FC, DIN, LC, MU, MTRJ, SC, ST, MPO, D4, Biconic, SMA, E-2000
- Simplex, duplex, and fanout (up to 24F) configurations; 0.9 mm, 2.0 mm, and 3.0 mm cable diameters
- Exchangeability and repeatability both ≤0.2 dB for consistent mating performance
- Singlemode fibers: OS2, G.657A1, G.657A2, multimode fibers: OM1, OM2, OM3, OM4
- Low insertion loss ≤0.3 dB (PC) for both SM and MM; high return loss ≥50 dB UPC, ≥60 dB APC
- Boot styles: standard, short boot, 90° bendable boot, and uniboot for high-density applications

Applications

- Data centre structured cabling and patch panel interconnections in high-density SM and MM environments
- Enterprise and campus LAN backbone and horizontal cabling with SM (OS2/G.657) or MM (OM3/OM4) fiber
- Telecommunications central office and outside plant cross-connects requiring low-loss, reliable patch cords
- FTTx access network distribution point and subscriber premises connections with bend-insensitive G.657A fiber

Technical Specifications

Parameter / Specification	Value / Detail
Cable Diameter	0.9 mm | 2.0 mm | 3.0 mm (or custom)
Cable Configuration	Simplex | Duplex | Fanout (up to 24F)
SM Fiber Type	OS2 | G.657A1 | G.657A2
MM Fiber Type	OM1 | OM2 | OM3 | OM4
Connector Types	FC | LC | SC | ST | MPO | MU | MTRJ | DIN | D4 | Biconic | SMA | E-2000
Polish	UPC | APC | PC
SM Insertion Loss (PC)	≤0.3 dB
SM Return Loss, UPC	≥50 dB
SM Return Loss, APC	≥60 dB
MM Insertion Loss (PC)	≤0.3 dB
MM Return Loss (PC)	≥35 dB
Exchangeability	≤0.2 dB
Repeatability	≤0.2 dB
Operating Temperature	−20°C to +75°C

TEST AND MEASUREMENT EQUIPMENT

Fusion Splicer PDR618H

Handheld core-alignment fusion splicer with fast splice and heating times.

The PDR618H is a compact handheld fusion splicer for SM, MM, DS, and NZDS fibers, offering fast splicing, quick heat shrinking, and automatic ARC calibration for reliable field and lab performance.

Applications

- Fiber-to-the-Home (FTTH) and Fiber-to-the-Building (FTTB) network installations
- Cable TV (CATV) and broadband infrastructure deployment
- Telecommunications network construction and maintenance
- Data centre interconnects and enterprise LAN cabling

Technical Specifications

Parameter / Specification	Value / Detail
Fiber Count	Single
Cladding Diameter	80 to 150 µm
Coating Diameter	160 to 900 µm
Typical Splice Loss	SM: 0.02 dB | MM: 0.01 dB | DS: 0.04 dB NZDS: 0.04 dB | G.657: 0.02 dB (ITU-T Standard)
Return Loss	> 60 dB
Fiber Cleaved Length	10 to 16 mm (coating dia. < 250 µm) 16 mm (coating dia. 250 to 900 µm)
Splicing Programs	40 groups
Operating Mode	Manual / Automatic
Auto-Heating	Available
Typical Splicing Time	8 seconds
Tube Heating Time	26 seconds (60 mm and 40 mm shrinkable sleeves)
Fiber View Magnification	250× (X or Y axis) | 125× (X and Y axis)
Viewing & Display	2× CMOS cameras, 4.3-inch colour LCD monitor
Splice Result Storage	4000 results

CABLE MANAGEMENT DEVICES

Heat Shrink Splice Closure

Sealed splice closure protecting fiber joints in outdoor and buried networks.

The PDR Fiber Optic Splice Closure is an IP68 weatherproof enclosure with heat shrink or re-enterable options, supporting up to 6 cable entries and 144 splices for outdoor fiber optic networks.

Key Features

- IP 68 sealing protection, suitable for fully immersed and buried deployments
- Zero additional insertion loss when fibers are routed through splice trays
- Unique strength member clamp assembly prevents cable sheath movement with temperature changes
- No specialized tools required, minimizes installation time and training costs
- Resistant to UV radiation and temperature fluctuation for deployment in all environments
- Bend controls in splicing tray guide and protect pigtail cables, ensuring controlled bending radius

Applications

- Fiber-to-the-Home (FTTH) and Fiber-to-the-Building (FTTB) outdoor splice points
- Aerial fiber optic cable splicing on poles and overhead installations
- Underground duct and direct-buried fiber optic cable joint protection
- Telecommunications network straight-through and branching joint applications

Technical Specifications

Parameter / Specification	Value / Detail
Dimension (L × D)	530 mm × 210 mm
Weight	< 2.7 kg
Type	Butt type
Cable Sealing	Heat shrink sleeves (Heat Shrink Type) | Mechanical seal (Mechanical Type)
Dome Material	PPCP

CABLE MANAGEMENT DEVICES

Horizontal Splice Closure

Horizontal splice closure for inline protection of fiber splices.

Re-enterable IP65 outdoor closure for duct, aerial, and wall mounting, supporting up to 48 splices (2×24) in a UV-resistant ribbed 445×240×140 mm body.

Key Features

- Suitable for cut, uncut, and taut sheath fiber optic cable applications
- Integrated sheath retention and central strength member fastening
- High resistance to chemical attack and UV degradation for extended outdoor life
- Compatible with all standard fiber optic cable types and diameters
- Ribbed closure construction using UV-resistant, lightweight plastic
- Supports up to 48 splices (2 × 24) with 16 SC simplex adapter capacity

Applications

- Fiber-to-the-Home (FTTH) and Fiber-to-the-Building (FTTB) feeder cable splicing
- Duct, aerial, and wall-mounted fiber optic cable distribution installations
- Telecommunications network outdoor splice points and cable junction management
- Cable TV (CATV) and broadband infrastructure outdoor splice enclosures

Technical Specifications

Parameter / Specification	Value / Detail
Model	HTSC-TL16
Dimensions (L×W×H)	445 × 240 × 140 mm
Splice Capacity	48 splices (2 × 24)
Main Cable Ports	4 ports, max 12 mm diameter
Drop Cable Ports	16 ports, 3 mm / 3×2 drop cable
Mounting Options	Duct, Aerial, Wall
Protection Rating	IP65

PASSIVE COMPONENTS

Hybrid Adapter

Hybrid fiber optic adapters that mate different connector types.

Mates two dissimilar connector types (SC-FC, LC-SC, ST-FC, etc.) via a precision ceramic/phosphor-bronze sleeve, ≤0.3 dB IL and ≥50 dB RL, for SM/MM fiber in simplex or duplex.

Key Features

- Enables connection between different connector types, SC-FC, LC-SC, LC-FC, ST-FC, LC-ST, and more
- Low insertion loss ≤ 0.3 dB and high return loss ≥ 50 dB (single-mode)
- Available in simplex and duplex configurations to suit varied cabling requirements
- High-precision ceramic or phosphor bronze alignment sleeve for superior optical performance
- Compatible with both single-mode (SM) and multi-mode (MM) fibers

Applications

- Fiber-to-the-Home (FTTH) and Fiber-to-the-Building (FTTB) network installations
- Cable TV (CATV) and broadband infrastructure interconnects
- Telecommunications network upgrades requiring connector-type migration
- Data center cross-connects and enterprise LAN cabling

PASSIVE COMPONENTS

Mode Conditioning Patchcord

Gigabit launch patchcord that runs single-mode laser sources over legacy multimode fiber without DMD.

Offset-launch patchcord that runs single-mode LX/LH laser sources over legacy 62.5/125 or 50/125 multimode fiber, eliminating Differential Mode Delay for reliable Gigabit Ethernet.

Key Features

- Ensures Gigabit transmission over multimode fiber
- Eliminates Differential Mode Delay (DMD)
- Offset launch technique from single-mode to multimode
- Compatible with 62.5/125 µm or 50/125 µm MMF cable plant
- Available with various connector types
- High return loss and low insertion loss
- High durability for repeated matings (≤ 0.20 dB typical change after 500 matings)

Applications

- Gigabit Ethernet (1000BASE-LX / LH) over legacy multimode fiber
- GBIC / SFP long-wavelength transceiver deployments on MMF cable plant
- Campus and building backbone links reusing existing multimode infrastructure
- Data centre and enterprise LAN interconnects

Technical Specifications

Parameter / Specification	Value / Detail
Cable Plant	62.5/125 or 50/125 Multimode Fiber (MMF)
Insertion Loss (MMF–SMF)	≤ 2.1 dB (50/125); ≤ 2.6 dB (62.5/125)
Return Loss	≥ 50 dB (UPC); ≥ 65 dB (APC)
Durability	≤ 0.20 dB typical after 500 matings
Operating Temperature	−20 °C to +75 °C

PASSIVE COMPONENTS

MPO Cable Assembly

High-density multi-fiber connector system

PDR MPO Cable Assemblies are high-density 4–24 fiber OS2/OM3/OM4/OM5 trunk cables with MT ferrules, ribbon or round cables, and male/female PC/APC connectors for fast data center deployment.

Key Features

- High-density MT ferrule with precision guide pins for reliable multi-fiber alignment in 4, 8, 12, and 24F configurations
- Fiber options: OS2 (singlemode), OM3, OM4, OM5 (multimode), supports 10GbE, 40GbE, and 100GbE migration
- Cable types: jacketed ribbon, bare ribbon, and round cable in single and double jacket constructions
- End-face polish: PC and APC; connector housings available in Green, Yellow, Beige, and Aqua colours
- Male and female connector options with push-pull latching for tool-free, secure mating
- SM insertion loss: 0.35 dB typical (standard), 0.20 dB typical (low loss); maximum 0.75 dB / 0.35 dB

Applications

- Data centre backbone and inter-rack cabling for 10GbE, 40GbE, and 100GbE high-density architectures
- MPO trunk cable distribution in structured cabling systems between main and horizontal distribution areas
- Telecom and datacom network infrastructure requiring rapid, high-density multi-fiber connectivity
- Optical fiber access networks and central office patch fields with large fiber counts

Technical Specifications

Parameter / Specification	Value / Detail
Fiber Mode	SM: OS2 | MM: OM3, OM4, OM5
Fiber Count	4F | 8F | 12F | 24F | 48F | 72F | 96F | 144F | Custom
Cable Type	Jacketed Ribbon | Bare Ribbon | Round
Cable Material	LSZH | PVC
Cable Diameter, Single Jacket 12F	3.0 mm
Cable Diameter, Double Jacket 12F/24F	4.5 mm
Tensile Strength, Single 12F (short/long)	200 N / 100 N
Tensile Strength, Double 12F/24F (short/long)	400 N / 150 N
Crush Resistance, Single 12F	500 N/100 mm
Crush Resistance, Double 12F	1,000 N/100 mm
Min. Bending Radius	10×D (installed) | 20×D (under load)
Cable Operating Temperature	, 20°C to +70°C
MPO Connector Type	Male or Female; PC or APC end face
Available Ferrule Counts	4, 8, 12, 24

PASSIVE COMPONENTS

Fiber Optic Cleaner Pen MPO

One-click cleaner pen for multi-fiber MPO/MTP connector end-faces.

PDR MPO/MTP Fiber Optic Cleaner Pen provides fast, dry cleaning of male and female MPO/MTP connectors, cleaning all 12 fibers in one push, fitting adapters and exposed ferrules, and delivering up to 500 cleaning cycles.

Key Features

- Pen-style handheld design with narrow profile for tightly-spaced MPO adapter panels
- Cleans all 12 fiber end-faces simultaneously in a single push-and-release action
- Dry mechanical cleaning, no alcohol or wet chemicals required
- Compatible with male (with guide pins) and female (without guide pins) MPO/MTP ferrules
- Effective removal of dust, oils, and particulate contaminants per IEC 61300-3-35
- Intermatability with FOCIS-5 (IEC 61754-7) compliant MPO connectors

Applications

- Data centre MPO/MTP patch panel and trunk cable maintenance
- Fiber-to-the-Home (FTTH) and Fiber-to-the-Building (FTTB) network installations
- Telecommunications network construction, testing, and field maintenance
- Cable TV (CATV) and broadband infrastructure connector cleaning

Technical Specifications

Parameter / Specification	Value / Detail
Compatible Connectors	MPO / MTP
Connector Standard	FOCIS-5 (IEC 61754-7)
Fiber Mode	Single-mode (SM) and Multimode (MM)
Ferrule Configuration	Male (with guide pins) and Female (without guide pins)
Polish Type	PC (Physical Contact) and APC (Angled Physical Contact)
Cleaning Method	Dry mechanical, no alcohol or solvents required
Fibers Cleaned per Stroke	12 fibers simultaneously
Cleaning Capacity	Up to 500 cleanings per unit
Operation	One-handed push-and-release mechanism
Cleaning Compatibility	Adapter-mounted connectors and exposed jumper ferrules
Contaminant Types	Dust, oils, fingerprints, and particulate matter
Cleaning Standard	IEC 61300-3-35 (fiber end-face cleanliness)
Design	Pen-style, handheld, narrow body for panel access

TEST AND MEASUREMENT EQUIPMENT

Next Gen Optical Fusion Splicer (OPM + VFL)

Next-generation fusion splicer with integrated power meter and visual fault locator.

The PDR4107S is a compact core-alignment fusion splicer for SM, MM, DS, and NZDS fibers, featuring 0.02 dB splice loss, a 5-second splice time, and built-in OPM and VFL for all-in-one field splicing.

Key Features

- Compact, lightweight design ideal for field installation and network maintenance
- Core alignment technology ensuring high-precision, low-loss fiber splicing
- Fast 5-second typical splicing time with automatic and manual operating modes
- Supports SM, MM, DS, and NZDS fiber types for maximum versatility
- Built-in Optical Power Meter (OPM), measurement range -50 to +26 dBm at 10 wavelengths
- Built-in Visual Fault Locator (VFL) with 10 mW output for fault identification

Applications

- Fiber-to-the-Home (FTTH) and Fiber-to-the-Building (FTTB) network installations
- Telecommunications network construction, deployment, and maintenance
- Cable TV (CATV) and broadband infrastructure rollout
- Data centre interconnects and enterprise LAN fiber cabling

Technical Specifications

Parameter / Specification	Value / Detail
Applicable Fibers	SM | MM | DS | NZDS
Fiber Count	Single
Cladding Diameter	80 to 150 µm
Coating Diameter	1 to 3 mm
Fiber Aligning Method	Automatic / Manual
Focusing Method	Four-motor automatic focusing
Typical Splice Loss	SM: 0.02 dB | MM: 0.01 dB | DS: 0.04 dB | NZDS: 0.04 dB
Return Loss	> 60 dB
Splicing Programs	53 preset + 40 user-defined
Splicing Modes	Auto / Calibrate / Normal / Special
Operating Mode	Automatic / Manual
Typical Splicing Time	5 seconds (standard SM fiber)
Tube Heating Programs	9 preset + 24 user-defined
Tube Heating Time	15 seconds | Protection sleeve: 60 mm / 40 mm

ACTIVE COMPONENTS > OPTICAL LINE PROTECTION SYSTEM (OLPS)

Optical Line Protection System (OLPS)

1U optical line protection with automatic switchover to a backup fiber path on failure.

PDR Optical Line Protection System (OLP): Compact 1U optical-layer protection with real-time monitoring, automatic fiber switching, 1+1/1:1/1-1 protection, redundant AC/DC power, and NMS support.

Key Features

- Compact 1U, 19-inch rack-mountable form factor
- Real-time optical power monitoring of both channels
- Configurable switching delay (0 to 999 s) and auto switchback
- Automatic fiber line switching on failure or degradation
- Three protection architectures: 1+1, 1:1 and 1-1
- Front-panel LCD with key control

Applications

- Telecom transmission network path redundancy
- Emergency rerouting of live traffic
- Real-time fiber quality surveillance
- FTTH/FTTB and access-network resilience

Technical Specifications

Parameter / Specification	Value / Detail
Protection Architecture	1+1 | 1:1 | 1-1
Form Factor	1U rack-mountable (19-inch)
Working Wavelength	1310 nm or 1550 nm (selectable)
Switching Threshold	Default −30 dBm (configurable)
Switching Delay	Immediate or 1 to 999 s
Auto Switchback	Configurable 1 to 999 minutes
Management	RJ-45 Ethernet | RS-232 DB9
Power Supply	220V AC | −48V DC (dual redundant)

PASSIVE COMPONENTS

PLC Splitter

Planar lightwave circuit splitters for low-loss PON power splitting.

PDR PLC Splitters offer passive optical power splitting over 1260–1650 nm in 1×N/2×N configurations with multiple packaging options for FTTx, PON, LAN, CATV, and test applications.

Key Features

- Planar waveguide chip technology for low insertion loss, high uniformity, and excellent long-term stability
- 1×N configurations: 1×2, 1×4, 1×8, 1×16, 1×32, 1×64
- 2×N configurations: 2×2, 2×4, 2×8, 2×16, 2×32, 2×64
- Wide operating wavelength: 1260 to 1650 nm, covers all standard PON and WDM bands
- Low polarization-dependent loss (PDL): ≤0.2 dB (1×2) to ≤0.4 dB (1×64)
- High directivity ≥55 dB and return loss ≥55 dB (1×N) / ≥50 dB (2×N)

TEST AND MEASUREMENT EQUIPMENT

Nano OTDR

Ultra-compact Nano OTDR with built-in VFL and power meter for quick field testing.

The PDR Nano OTDR is a compact, smartphone-connected dual-wavelength (1310/1550 nm) OTDR for precise fiber testing, loss analysis, and real-time measurements via the Smart-OTDR app.

Key Features

- Ultra-compact nano size, fits in a shirt pocket for true field portability
- Dual-wavelength operation at 1310 nm and 1550 nm for comprehensive fiber testing
- Wireless Bluetooth / Wi-Fi connectivity to the Smart-OTDR smartphone app (Android & iOS)
- Real-time OTDR trace display, zoom, and cursor analysis on smartphone screen
- Measures Total Loss, Total ORL, Average Loss, and Distance per event automatically
- Multiple pulse widths for optimised dead-zone and dynamic-range trade-off

Applications

- Fiber-to-the-Home (FTTH) and Fiber-to-the-Building (FTTB) acceptance testing
- Telecommunications network construction, commissioning, and maintenance
- Cable TV (CATV) and broadband passive optical network (PON) verification
- Data centre interconnect fiber certification and fault localisation

Technical Specifications

Parameter / Specification	Value / Detail
Fiber Type	Single-Mode (ITU-T G.652 / G.657)
Wavelengths	1310 nm | 1550 nm
Dynamic Range	26 dB @ 1310 nm | 24 dB @ 1550 nm
Distance Range	0.1 km, 60 km (auto & manual)
Distance Resolution	0.01 m
Distance Accuracy	±1 m + ±0.005% × distance
Event Dead Zone	≤ 1 m (typical, with 10 ns pulse)
Attenuation Dead Zone	≤ 4 m (typical, with 10 ns pulse)
Pulse Widths	10 ns / 30 ns / 100 ns / 300 ns / 1 µs / 3 µs
Sampling Resolution	4 cm
Loss Threshold	0.01 dB resolution
Return Loss (ORL)	Measured in dB, displayed in app
Averaging Time	5 s / 15 s / 30 s / 60 s / 180 s
Connectivity	Bluetooth 4.0 | Wi-Fi 802.11 b/g/n

PASSIVE COMPONENTS

POF — Plastic Optical Fiber Patchcord

Large PMMA core for easy alignment

PDR Plastic Optical Fiber (POF) Cable Assemblies provide cost-effective, EMI-immune connectivity with SI-POF cables, LC/RedLink™/VersaLink® patch cords, reel options for custom installation, UL VW-1 jackets, and compatibility with standard POF transceivers.

Key Features

- 1 mm step-index SI-POF with numerical aperture 0.5, compliant with IEC 60793-2-40 Class A4a
- Simplex cable OD 2.2 mm; duplex (figure-of-8) minor axis 2.2 mm, major axis 4.4 mm
- Minimum bend radius 25 mm; repeat bending endurance ≥10,000 cycles (≤1 dB additional loss)
- Low attenuation: 0.19 dB/m at 650 nm, ideal for short-range links up to tens of metres
- UL VW-1 flame-retardant jacket option available for building and equipment installations
- Tensile rating: 70 N simplex / 140 N duplex cable

Applications

- Control links within high-voltage electrical control equipment requiring full galvanic isolation
- Industrial automation links between equipment requiring maintained electrical isolation
- Data communications systems in environments with extreme electromagnetic interference (EMI)
- Rugged short-range optical links in harsh factory, process control, and automotive environments

Technical Specifications

Parameter / Specification	Value / Detail
Fiber Standard	IEC 60793-2-40 Class A4a (SI-POF)
Numerical Aperture	0.5
Attenuation at 650 nm	≤0.19 dB/m
Fiber Diameter (incl. cladding)	0.94 to 1.06 mm (nominal 1.0 mm)
Cable Diameter, Simplex	2.13 to 2.27 mm (nominal 2.2 mm)
Cable Diameter, Duplex Minor Axis	2.13 to 2.27 mm
Cable Diameter, Duplex Major Axis	4.3 to 4.5 mm
Jacket Material	Flame-resistant PE or PVC
Jacket Color	Black; Duplex cable has white identifying bar on one strand
UL Flame Rating	VW-1
Tensile Stress, Simplex / Duplex	70 N / 140 N
Minimum Bend Radius	25 mm
Repeat Bending Endurance	≥10,000 cycles (≤1 dB additional loss)
Operating Temperature	-40°C to +85°C

TEST AND MEASUREMENT EQUIPMENT

PON Power Meter

PON power meter with pass-through measurement for live FTTx networks.

The PDR4213B is a handheld PON Power Meter for simultaneous 1310/1490/1550 nm measurement, providing fast, accurate FTTH/GPON/EPON testing for installation and maintenance.

Applications

- FTTH / FTTB installation acceptance testing and power budget verification
- GPON and EPON network commissioning, maintenance, and fault localisation
- Simultaneous upstream (1310 nm) and downstream (1490 / 1550 nm) signal verification
- PON splitter loss measurement and passive component characterisation

Technical Specifications

Parameter / Specification	Value / Detail
Wavelengths	1310 nm | 1490 nm | 1550 nm
Measurement Mode	Simultaneous 3-channel power measurement
Power Range	-50 dBm to +26 dBm
Resolution	0.01 dB | 0.001 mW
Accuracy	+/- 0.25 dB (at reference wavelength, 23 C)
Display Units	dBm | dBr | mW (user selectable)
Detector Type	InGaAs photodiode
Optical Connector	SC/APC (SC/UPC adapter included)
Connector Interface	SC adapter port with 2.5 mm ferrule
Data Storage	Up to 99 measurement records (SAVE function)
Data Interface	USB 2.0 (data transfer and firmware upgrade)
Battery	3 x AA alkaline batteries
Battery Life	>= 40 hours continuous operation
Auto Power-Off	Configurable: 5 / 10 / 15 min or disabled

CABLE MANAGEMENT DEVICES

Rack Mount Fiber Management System

Modular 19-inch rack-mount fiber management system for high-density termination.

PDR 19-inch Rack Mount Fiber Management System provides modular, scalable, high-density fiber management in durable 1U/2U/4U CRCA steel enclosures with sliding tray access, supporting 6–192 fiber terminations.

Key Features

- 19-inch rack mount enclosure available in 1U, 2U, and 4U heights for scalable fiber density
- Supports 6F to 192F fiber terminations per enclosure, FC, SC, ST, and LC connector types
- Removable top cover and sliding tray mechanism for quick and easy access to service connections
- Multiple cable entry and exit openings on all models for flexible routing and installation
- Slide-out mechanism provides excellent cable management and organizes cable slack efficiently
- Maintains minimum bend radius of pigtails and patch cords to protect fiber integrity

Applications

- Data centre main distribution frames (MDF) and structured cabling infrastructure
- Telecommunications exchange and central office fiber termination panels
- Enterprise LAN fiber backbone and horizontal cabling patch points
- FTTH and FTTB aggregation points in multi-dwelling unit (MDU) installations

Technical Specifications

Parameter / Specification	Value / Detail
Rack Compatibility	19-inch standard rack mount
Available Rack Heights	1U | 2U | 4U
Fiber Capacity Range	6F to 192F per enclosure
Connector Types	FC / SC / ST (XX series) | LC / PC (LC series)
Material	CRCA (Cold-Rolled Close-Annealed steel)
Finish	Structure finish and powder coated, 80 to 100 µm
Colour	RAL 7032 (standard) | Customised on request
Cable Entry	Rear
Cable Mounting	Horizontal
Splice Tray	Hinged type ABS
Top Cover	Removable for service access
Access Mechanism	Sliding tray
Rodent Cover	Available (optional)
Dimensions (1U models)	485 mm × 315 mm × 45 mm

PASSIVE COMPONENTS

FTTX Smart Bullet Drop Cable

FTTXSmart pre-connectorized bullet drop cable for fast last-mile FTTH deployment.

Single-fiber SC/APC drop assembly with a Ø5.9 mm bullet head for traction pulling through narrow ducts, ≤0.3 dB IL, ≥60 dB RL over 1260–1650 nm, 100% optically tested.

Key Features

- Bullet-shaped connector head (Ø5.9 mm) with guide-wire hole for traction installation through narrow ducts
- SC/APC connectors with zirconia ceramic ferrule: insertion loss ≤0.3 dB (typ. 0.2 dB), return loss ≥60 dB
- APC ferrule angle 8° ±0.3° for superior back-reflection performance in FTTH and PON networks
- Pull-through design for free and occupied ducts, no large bore holes, conduits, or specialist tools required
- Wide operating wavelength 1260 to 1650 nm covers all standard telecom bands including O, E, S, C, L, and U
- Mechanical durability ≥500 insertion/removal cycles; repeatability and interchangeability ≤0.1 dB

Technical Specifications

Parameter / Specification	Value / Detail
Connector Type	SC/APC bullet-shaped (near-end) / SC/APC (far-end)
Bullet Connector Diameter	Ø5.9 mm
Operating Wavelength	1260 to 1650 nm
Insertion Loss	≤0.3 dB (typ. 0.2 dB)
Return Loss	≥60 dB
Tractive Force	≥150 N
Working Temperature	−40°C to +70°C

TEST AND MEASUREMENT EQUIPMENT > POWER METER

Mini Optical Power Meter

Palm-sized handheld optical power meter for FTTH continuity and link-loss testing.

Palm-sized handheld meter for FTTH continuity and link-loss testing, six calibrated wavelengths (850–1625 nm), ±0.2 dB accuracy, 0.01 dB resolution, SM/MM via 2.5 mm connector, up to 100 h battery.

Key Features

- Palm-sized, pocket-carry handheld design for one-hand field use
- Six calibrated wavelengths: 850 / 1300 / 1310 / 1490 / 1550 / 1625 nm
- Optional visual fault locator (1 mW or 10 mW, CW and glint) and LED SOS output
- Battery autonomy above 100 hours; runs on 2× AA or AC/DC adapter
- Wide measurement range with high 0.01 dB resolution and ±0.2 dB accuracy
- Supports single-mode and multimode fiber via 2.5 mm universal connector
- Auto power-off after 10 minutes (cancellable) to conserve battery
- One-year warranty with three-year recommended calibration interval

Applications

- FTTH in-house installation and continuity testing
- Optical fiber network installation and maintenance
- Fiber link transmission quality evaluation
- Single-mode and multimode fiber measurement
- Field service and troubleshooting by technicians

Technical Specifications

Parameter / Specification	Value / Detail
Display Range (PDR60T)	+6 to −70 dBm (1310/1490/1550/1625 nm)
Display Range (PDR60C)	+26 to −50 dBm (1310/1490/1550/1625 nm)
Accuracy	±0.2 dB
Calibrated Wavelengths	850 / 1300 / 1310 / 1490 / 1550 / 1625 nm
Display Resolution	0.01 dB
Battery Life	Above 100 hours
Power Supply	2× AA or AC/DC adapter
Dimensions / Weight	105 × 52 × 34 mm; ~100 g

ACTIVE COMPONENTS

100G & 100G BIDI

Full range of MSA-compliant transceivers from 1.25G to 400G across SFP and QSFP form factors.

PDR optical transceivers deliver hot-pluggable, MSA-compliant 1.25G–400G connectivity with SFP to QSFP-DD form factors, DDM, and SM/MM, CWDM/DWDM/BiDi support.

Key Features

- MSA-compliant form factors: SFP, SFP+, SFP28, QSFP+, QSFP28, QSFP-DD and Copper SFP
- Aggregate data rates from 1.25 Gbps to 400 Gbps
- Single-mode and multimode options with reach up to 80 km
- Hot-pluggable with duplex LC (SFP) and MPO/MTP (QSFP) receptacles
- Digital Diagnostic Monitoring (DDM/DOM) on optical variants
- CWDM, DWDM and BiDi variants available on request

Applications

- Telecom backbone, metro Ethernet and long-haul DWDM
- Data-centre server-to-switch and spine aggregation
- Enterprise and campus LAN uplinks

Technical Specifications

Parameter / Specification	Value / Detail
SFP	Up to 2.67 Gbps · SM/MM · up to 80 km
SFP+	Up to 11.3 Gbps · SM/MM · up to 40 km
SFP28	25.78 Gbps · SM/MM · up to 10 km
QSFP+ (40G)	41.2 Gbps (4×10G) · up to 10 km
QSFP28 (100G)	103.1 Gbps (4×25G) · up to 2 km
QSFP-DD (200G/400G)	200G / 400G · up to 2 km
Copper SFP	10/100/1000BASE-T · Cat 5 · 100 m
Operating Temperature	0°C to 70°C
Compliance	Relevant MSA, IEEE 802.3, RoHS/REACH

ACTIVE COMPONENTS

SFP+ 10G BIDI

Full range of MSA-compliant transceivers from 1.25G to 400G across SFP and QSFP form factors.

PDR optical transceivers deliver hot-pluggable, MSA-compliant 1.25G–400G connectivity with SFP to QSFP-DD form factors, DDM, and SM/MM, CWDM/DWDM/BiDi support.

Key Features

- MSA-compliant form factors: SFP, SFP+, SFP28, QSFP+, QSFP28, QSFP-DD and Copper SFP
- Aggregate data rates from 1.25 Gbps to 400 Gbps
- Single-mode and multimode options with reach up to 80 km
- Hot-pluggable with duplex LC (SFP) and MPO/MTP (QSFP) receptacles
- Digital Diagnostic Monitoring (DDM/DOM) on optical variants
- CWDM, DWDM and BiDi variants available on request

Applications

- Telecom backbone, metro Ethernet and long-haul DWDM
- Data-centre server-to-switch and spine aggregation
- Enterprise and campus LAN uplinks

Technical Specifications

Parameter / Specification	Value / Detail
SFP	Up to 2.67 Gbps · SM/MM · up to 80 km
SFP+	Up to 11.3 Gbps · SM/MM · up to 40 km
SFP28	25.78 Gbps · SM/MM · up to 10 km
QSFP+ (40G)	41.2 Gbps (4×10G) · up to 10 km
QSFP28 (100G)	103.1 Gbps (4×25G) · up to 2 km
QSFP-DD (200G/400G)	200G / 400G · up to 2 km
Copper SFP	10/100/1000BASE-T · Cat 5 · 100 m
Operating Temperature	0°C to 70°C
Compliance	Relevant MSA, IEEE 802.3, RoHS/REACH

ACTIVE COMPONENTS

SFP+ 10G Dual Fiber

Standard dual-fiber 10G transmission

PDR optical transceivers deliver hot-pluggable, MSA-compliant 1.25G–400G connectivity with SFP to QSFP-DD form factors, DDM, and SM/MM, CWDM/DWDM/BiDi support.

Key Features

- MSA-compliant form factors: SFP, SFP+, SFP28, QSFP+, QSFP28, QSFP-DD and Copper SFP
- Aggregate data rates from 1.25 Gbps to 400 Gbps
- Single-mode and multimode options with reach up to 80 km
- Hot-pluggable with duplex LC (SFP) and MPO/MTP (QSFP) receptacles
- Digital Diagnostic Monitoring (DDM/DOM) on optical variants
- CWDM, DWDM and BiDi variants available on request

Applications

- Telecom backbone, metro Ethernet and long-haul DWDM
- Data-centre server-to-switch and spine aggregation
- Enterprise and campus LAN uplinks

Technical Specifications

Parameter / Specification	Value / Detail
SFP	Up to 2.67 Gbps · SM/MM · up to 80 km
SFP+	Up to 11.3 Gbps · SM/MM · up to 40 km
SFP28	25.78 Gbps · SM/MM · up to 10 km
QSFP+ (40G)	41.2 Gbps (4×10G) · up to 10 km
QSFP28 (100G)	103.1 Gbps (4×25G) · up to 2 km
QSFP-DD (200G/400G)	200G / 400G · up to 2 km
Copper SFP	10/100/1000BASE-T · Cat 5 · 100 m
Operating Temperature	0°C to 70°C
Compliance	Relevant MSA, IEEE 802.3, RoHS/REACH

ACTIVE COMPONENTS

1G SFP BIDI

Cost-effective 1G bi-directional link

PDR optical transceivers deliver hot-pluggable, MSA-compliant 1.25G–400G connectivity with SFP to QSFP-DD form factors, DDM, and SM/MM, CWDM/DWDM/BiDi support.

Key Features

- MSA-compliant form factors: SFP, SFP+, SFP28, QSFP+, QSFP28, QSFP-DD and Copper SFP
- Aggregate data rates from 1.25 Gbps to 400 Gbps
- Single-mode and multimode options with reach up to 80 km
- Hot-pluggable with duplex LC (SFP) and MPO/MTP (QSFP) receptacles
- Digital Diagnostic Monitoring (DDM/DOM) on optical variants
- CWDM, DWDM and BiDi variants available on request

Applications

- Telecom backbone, metro Ethernet and long-haul DWDM
- Data-centre server-to-switch and spine aggregation
- Enterprise and campus LAN uplinks

Technical Specifications

Parameter / Specification	Value / Detail
SFP	Up to 2.67 Gbps · SM/MM · up to 80 km
SFP+	Up to 11.3 Gbps · SM/MM · up to 40 km
SFP28	25.78 Gbps · SM/MM · up to 10 km
QSFP+ (40G)	41.2 Gbps (4×10G) · up to 10 km
QSFP28 (100G)	103.1 Gbps (4×25G) · up to 2 km
QSFP-DD (200G/400G)	200G / 400G · up to 2 km
Copper SFP	10/100/1000BASE-T · Cat 5 · 100 m
Operating Temperature	0°C to 70°C
Compliance	Relevant MSA, IEEE 802.3, RoHS/REACH

ACTIVE COMPONENTS

25G & 25G BIDI

Optimized for 5G fronthaul

PDR optical transceivers deliver hot-pluggable, MSA-compliant 1.25G–400G connectivity with SFP to QSFP-DD form factors, DDM, and SM/MM, CWDM/DWDM/BiDi support.

Key Features

- MSA-compliant form factors: SFP, SFP+, SFP28, QSFP+, QSFP28, QSFP-DD and Copper SFP
- Aggregate data rates from 1.25 Gbps to 400 Gbps
- Single-mode and multimode options with reach up to 80 km
- Hot-pluggable with duplex LC (SFP) and MPO/MTP (QSFP) receptacles
- Digital Diagnostic Monitoring (DDM/DOM) on optical variants
- CWDM, DWDM and BiDi variants available on request

Applications

- Telecom backbone, metro Ethernet and long-haul DWDM
- Data-centre server-to-switch and spine aggregation
- Enterprise and campus LAN uplinks

Technical Specifications

Parameter / Specification	Value / Detail
SFP	Up to 2.67 Gbps · SM/MM · up to 80 km
SFP+	Up to 11.3 Gbps · SM/MM · up to 40 km
SFP28	25.78 Gbps · SM/MM · up to 10 km
QSFP+ (40G)	41.2 Gbps (4×10G) · up to 10 km
QSFP28 (100G)	103.1 Gbps (4×25G) · up to 2 km
QSFP-DD (200G/400G)	200G / 400G · up to 2 km
Copper SFP	10/100/1000BASE-T · Cat 5 · 100 m
Operating Temperature	0°C to 70°C
Compliance	Relevant MSA, IEEE 802.3, RoHS/REACH

ACTIVE COMPONENTS

400G

Next-generation high-capacity throughput

PDR optical transceivers deliver hot-pluggable, MSA-compliant 1.25G–400G connectivity with SFP to QSFP-DD form factors, DDM, and SM/MM, CWDM/DWDM/BiDi support.

Key Features

- MSA-compliant form factors: SFP, SFP+, SFP28, QSFP+, QSFP28, QSFP-DD and Copper SFP
- Aggregate data rates from 1.25 Gbps to 400 Gbps
- Single-mode and multimode options with reach up to 80 km
- Hot-pluggable with duplex LC (SFP) and MPO/MTP (QSFP) receptacles
- Digital Diagnostic Monitoring (DDM/DOM) on optical variants
- CWDM, DWDM and BiDi variants available on request

Applications

- Telecom backbone, metro Ethernet and long-haul DWDM
- Data-centre server-to-switch and spine aggregation
- Enterprise and campus LAN uplinks

Technical Specifications

Parameter / Specification	Value / Detail
SFP	Up to 2.67 Gbps · SM/MM · up to 80 km
SFP+	Up to 11.3 Gbps · SM/MM · up to 40 km
SFP28	25.78 Gbps · SM/MM · up to 10 km
QSFP+ (40G)	41.2 Gbps (4×10G) · up to 10 km
QSFP28 (100G)	103.1 Gbps (4×25G) · up to 2 km
QSFP-DD (200G/400G)	200G / 400G · up to 2 km
Copper SFP	10/100/1000BASE-T · Cat 5 · 100 m
Operating Temperature	0°C to 70°C
Compliance	Relevant MSA, IEEE 802.3, RoHS/REACH

ACTIVE COMPONENTS

40G

Reliable mid-tier aggregation

PDR optical transceivers deliver hot-pluggable, MSA-compliant 1.25G–400G connectivity with SFP to QSFP-DD form factors, DDM, and SM/MM, CWDM/DWDM/BiDi support.

Key Features

- MSA-compliant form factors: SFP, SFP+, SFP28, QSFP+, QSFP28, QSFP-DD and Copper SFP
- Aggregate data rates from 1.25 Gbps to 400 Gbps
- Single-mode and multimode options with reach up to 80 km
- Hot-pluggable with duplex LC (SFP) and MPO/MTP (QSFP) receptacles
- Digital Diagnostic Monitoring (DDM/DOM) on optical variants
- CWDM, DWDM and BiDi variants available on request

Applications

- Telecom backbone, metro Ethernet and long-haul DWDM
- Data-centre server-to-switch and spine aggregation
- Enterprise and campus LAN uplinks

Technical Specifications

Parameter / Specification	Value / Detail
SFP	Up to 2.67 Gbps · SM/MM · up to 80 km
SFP+	Up to 11.3 Gbps · SM/MM · up to 40 km
SFP28	25.78 Gbps · SM/MM · up to 10 km
QSFP+ (40G)	41.2 Gbps (4×10G) · up to 10 km
QSFP28 (100G)	103.1 Gbps (4×25G) · up to 2 km
QSFP-DD (200G/400G)	200G / 400G · up to 2 km
Copper SFP	10/100/1000BASE-T · Cat 5 · 100 m
Operating Temperature	0°C to 70°C
Compliance	Relevant MSA, IEEE 802.3, RoHS/REACH

ACTIVE COMPONENTS

Copper SFP+

RJ-45 interface for twisted pair copper

PDR optical transceivers deliver hot-pluggable, MSA-compliant 1.25G–400G connectivity with SFP to QSFP-DD form factors, DDM, and SM/MM, CWDM/DWDM/BiDi support.

Key Features

- MSA-compliant form factors: SFP, SFP+, SFP28, QSFP+, QSFP28, QSFP-DD and Copper SFP
- Aggregate data rates from 1.25 Gbps to 400 Gbps
- Single-mode and multimode options with reach up to 80 km
- Hot-pluggable with duplex LC (SFP) and MPO/MTP (QSFP) receptacles
- Digital Diagnostic Monitoring (DDM/DOM) on optical variants
- CWDM, DWDM and BiDi variants available on request

Applications

- Telecom backbone, metro Ethernet and long-haul DWDM
- Data-centre server-to-switch and spine aggregation
- Enterprise and campus LAN uplinks

Technical Specifications

Parameter / Specification	Value / Detail
SFP	Up to 2.67 Gbps · SM/MM · up to 80 km
SFP+	Up to 11.3 Gbps · SM/MM · up to 40 km
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QSFP+ (40G)	41.2 Gbps (4×10G) · up to 10 km
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QSFP-DD (200G/400G)	200G / 400G · up to 2 km
Copper SFP	10/100/1000BASE-T · Cat 5 · 100 m
Operating Temperature	0°C to 70°C
Compliance	Relevant MSA, IEEE 802.3, RoHS/REACH

ACTIVE COMPONENTS

Smart SFP Transceiver

Integrated OAM (Operations, Administration, and Maintenance)

Integrated OAM (Operations, Administration, and Maintenance)

Key Features

- Integrated OAM (Operations, Administration, and Maintenance)
- Micro-loopback capability
- Real-time performance monitoring

Applications

- Service Provider Edge
- SLA Monitoring
- Mobile Backhaul

Technical Specifications

Parameter / Specification	Value / Detail
Interface	1G / 10G SFP
Protocol	IEEE 802.3ah / 802.1ag
Diagnostics	DDM/DOM
Latency	Ultra-Low

PASSIVE COMPONENTS

SMPTE Cable Assembly

Ruggedized, dirt-protected fiber assemblies engineered for HDTV broadcast infrastructure.

Ruggedized, dirt-protected SM/MM fiber assemblies with all-metal housings, heavy-duty retention, and silicon gaskets, engineered for HDTV broadcast and outdoor OB applications

Key Features

- Rugged all-metal connector housing for durability
- Silicon gasket protection against dust and dirt
- Supports single-mode (9/125 µm) and multimode (50/125 µm, 62.5/125 µm, OM3) fiber
- Customizable lengths from 0.5 m to 999 m
- Heavy-duty cable retention for secure long-term performance
- Wide connector range: SC, LC, FC, ST, MU, MTRJ, SMA, E2000
- Multiple jacket types: simplex, duplex, mini-zipcord, armored, FTTH

Applications

- HDTV broadcast infrastructure
- Outdoor field deployment and Outside Broadcast (OB) vans System integration
- Broadcast studio installations

Technical Specifications

Parameter / Specification	Value / Detail
Fiber Type	Single-mode (9/125 µm); Multimode (50/125, 62.5/125, OM3)
Connector Options	SC, LC, FC, ST, MU, MTRJ, SMA, E2000
Insertion Loss	≤ 0.30 dB typical; ≤ 0.50 dB max
Return Loss	≥ 50 dB UPC; ≥ 60 dB APC
Length Options	0.5 m to 999 m (custom available)
Operating Temperature	−40 °C to +75 °C

PASSIVE COMPONENTS

VFL — Visual Fault Locator

Visual fault locator for pinpointing fiber breaks, bends and faults.

The BML209 is a compact pen-style 650 nm Visual Fault Locator (VFL) that detects fiber faults, breaks, bends, and connector defects, with 1–30 mW output options for 3 km to 10+ km testing.

Key Features

- 650 nm ±20 nm visible red laser for instant fault and break localisation
- Four output power variants: 1 mW, 10 mW, 20 mW, and 30 mW for different reach requirements
- Dynamic detection distance: 3 to 5 km (1 mW) up to >10 km (20/30 mW variants)
- Dual operation modes: Continuous Wave (CW) and Pulsed (2 Hz) for identification
- 2.5 mm universal adapter connector; optional 1.25 mm adapter available
- Detects faults in connectors, splices, bends, breaks, and fiber splitter points

Technical Specifications

Parameter / Specification	Value / Detail
Output Power	>1 mW / >10 mW / >20 mW / >30 mW
Dynamic Distance	3 to 5 km / 8 to 10 km / >10 km / >10 km
Wavelength	650 nm ±20 nm
Operation Modes	Continuous Wave (CW) and Pulsed (2 Hz)
Connector	2.5 mm Universal adapter
Dimensions	180 mm (L) × 30 mm (diameter)
Weight	46 g

CABLE MANAGEMENT DEVICES

Optical Fiber Wall Mount Enclosure

Wall-mount optical enclosure for protected indoor fiber termination.

PDR Optical Fiber Wall Mount Enclosure is a lockable double-door fiber termination unit for Singlemode/Multimode networks, supporting ST, FC, SC & LC adapters, loaded or unloaded options, and four cable entry points for flexible installation.

Key Features

- Scalable port capacity, available in 12, 24, 48, and 96 port configurations
- Double-door construction with key lock for secure, access-controlled fiber termination
- Four-point cable entry (two top, two bottom) for flexible fiber routing
- Accepts Singlemode and Multimode ST, FC, SC, and LC adapter plates
- Supplied loaded (with adapters) or unloaded, installer configures as required
- Compact wall-mount footprint eliminates the need for a full 19" cabinet

Applications

- Fiber-to-the-Home (FTTH) and Fiber-to-the-Building (FTTB) premises termination
- Telecommunications network distribution points and cross-connect locations
- Enterprise LAN and structured cabling fiber termination in server rooms
- CATV and broadband fiber distribution in residential and commercial buildings

ACTIVE COMPONENTS

Bypass Switch

Carrier-grade 1U optical bypass switch that reroutes traffic around a failed inline device in under 50 ms.

Carrier-grade 1U optical path protection that reroutes traffic around a failed inline device in under 50 ms via power, optical-power, and TCP/IP heartbeat monitoring, with up to 4 channels and dual AC/DC power.

Key Features

- Automatic bypass protection via power supply, optical power, and TCP/IP heartbeat monitoring — individually or in combination
- Sub-50 ms carrier-grade switching speed for transparent network failover
- Up to 4 independent bypass channels in a single 1U 19-inch rack chassis
- Auto-Switch and Manual-Switch modes with configurable auto-return after an idle period
- LC/PC single-mode (9/125 µm) interfaces with ≤ 1.5 dB insertion loss and ≥ 50 dB return loss
- Switching-log records reason, time, and result of every switching event
- Dual power inputs: AC 85–264 V (50/60 Hz) and DC 36–72 V for full redundancy

Applications

- Inline node protection for network security appliances — firewalls, IPS, DPI, and flow-control devices
- FTTH/FTTB access network deployments
- Military, industrial, and mission-critical communication systems
- Repeater and optical amplifier protection in long-haul and metro networks
- Data centre interconnects and spine-leaf fabric resilience

Technical Specifications

Parameter / Specification	Value / Detail
Working Wavelength	1310 nm / 1550 nm (selectable)
Fiber Type	SM 9/125 µm (ITU-T G.652)
Channels	Up to 4 independent channels
Insertion Loss	≤ 1.5 dB
Return Loss	≥ 50 dB
Switching Time	< 50 ms
Power Supply	AC 85–264 V | DC 36–72 V (dual inputs)
Dimensions	19-inch rack, 1U height

CABLE MANAGEMENT DEVICES

Home Termination Box (HTB)

Compact home termination box for neat FTTH subscriber-side fiber management.

PDR Home Termination Box (HTB) is a compact wall-mounted enclosure for fiber termination, splicing, adapters, FICs, pigtails and cable storage, with bend-radius protection and splice/slack management for reliable ONT/CPE connections.

Key Features

- Compact, low-profile design occupies minimal wall space during installation
- Inner slack storage area maintains minimum bend radius compliance at all times
- Simple, tool-free installation with included mounting screws and cable ties
- Inner tray accommodates splice sleeves for clean in-box splicing
- Accepts both standard SC and LC optical adapters
- Available in Loaded (with adapters) and Unloaded (adapter-free) configurations

Applications

- Fiber-to-the-Home (FTTH) and Fiber-to-the-Building (FTTB) last-mile deployments
- Optical CATV and broadband fiber distribution networks
- Local Area Network (LAN) structured cabling termination points
- ONT patch cord interface in residential and small commercial premises

Technical Specifications

Parameter / Specification	Value / Detail
Dimensions (L×W×H)	86 × 86 × 25 mm
Fiber Count	2 Fiber
Colour	White
Material	ABS, PP
Insertion Loss	< 0.3 dB
Return Loss	≥ 65 dB (APC) / ≥ 55 dB (UPC)
Adapter Compatibility	SC, LC (standard form factor)
Configuration	Loaded / Unloaded
Standard	IEC 61754-4

ACTIVE COMPONENTS

1G SFP Dual Fiber

Legacy compatible 1G module

PDR optical transceivers deliver hot-pluggable, MSA-compliant 1.25G–400G connectivity with SFP to QSFP-DD form factors, DDM, and SM/MM, CWDM/DWDM/BiDi support.

Key Features

- MSA-compliant form factors: SFP, SFP+, SFP28, QSFP+, QSFP28, QSFP-DD and Copper SFP
- Aggregate data rates from 1.25 Gbps to 400 Gbps
- Single-mode and multimode options with reach up to 80 km
- Hot-pluggable with duplex LC (SFP) and MPO/MTP (QSFP) receptacles
- Digital Diagnostic Monitoring (DDM/DOM) on optical variants
- CWDM, DWDM and BiDi variants available on request

Applications

- Telecom backbone, metro Ethernet and long-haul DWDM
- Data-centre server-to-switch and spine aggregation
- Enterprise and campus LAN uplinks

Technical Specifications

Parameter / Specification	Value / Detail
SFP	Up to 2.67 Gbps · SM/MM · up to 80 km
SFP+	Up to 11.3 Gbps · SM/MM · up to 40 km
SFP28	25.78 Gbps · SM/MM · up to 10 km
QSFP+ (40G)	41.2 Gbps (4×10G) · up to 10 km
QSFP28 (100G)	103.1 Gbps (4×25G) · up to 2 km
QSFP-DD (200G/400G)	200G / 400G · up to 2 km
Copper SFP	10/100/1000BASE-T · Cat 5 · 100 m
Operating Temperature	0°C to 70°C
Compliance	Relevant MSA, IEEE 802.3, RoHS/REACH

SPECIALTY DRONES

UAV Fiber Optic Spool

Ultra-light UAV payout spool that dispenses bend-insensitive fiber cleanly at high speed, from 500 m to 50 km.

Ultra-light optical fiber payout spool for tethered UAVs and unmanned vehicles, wound with bend-insensitive G.657.A2 single-mode fiber supplied in 17 standard lengths (500 m–50 km) with an FC interface.

Key Features

- Ultra-light payout spool design optimized for tethered UAV and unmanned-vehicle fiber links
- Bend-insensitive G657.A2 single-mode fiber for reliable high-speed payout with minimal bend loss
- Optional Kevlar coating for high strength and wear resistance in harsh environments
- Rated 60 N pulling force for controlled, snag-free fiber dispensing under tension
- Rugged black housing built for demanding field and airborne conditions
- Made in India

Applications

- Tethered UAV and drone fiber-optic communication and control links
- Defence, ISR and loitering / FPV unmanned systems requiring jam-resistant fiber tethers
- Unmanned surface and underwater vehicles (USV/ROV) and unmanned vessels
- Underwater exploration, tunnels, pipelines and underground culvert deployment
- Rapid field-deployable and emergency fiber-optic communication links
- Industrial and research applications demanding lightweight, high-strength fiber payout

Technical Specifications

Parameter / Specification	Value / Detail
Weight	80 g/km
Pulling Force	60N
Available Lengths	500 m to 50 km (17 standard variants — see Ordering Information)

SPECIALTY DRONES

Sky Unit

Airborne fiber optical terminal that sends camera video and flight-controller telemetry down a single fiber at 1.25 Gbps.

Aircraft-mounted transmit terminal that accepts camera composite video plus flight-controller telemetry and carries them over a single fiber at 1.25 Gbps, offered in four compact mounting sizes with a GH1.25 interface for direct integration into the drone's power and control harness.

Key Features

- Airborne (Type A) transmit terminal for tethered / FPV drone fiber links
- Accepts composite video input and flight-controller TTL telemetry
- GH1.25 (8P) back-panel connector carrying video, TTL serial and power
- Wide 5–12 V DC input for direct drone-power integration
- Single-fiber optical transmission at 1.25 Gbps over an FC interface
- Four mounting sizes (88×88 to 141×120 mm) for flexible airframe integration
- PWR and SYC status LEDs for power and optical-sync diagnostics

Applications

- Airborne video/telemetry transmission for FPV and racing drones
- Long-range ISR and defence UAV downlinks
- Tethered-drone and loitering unmanned systems needing jam-resistant links
- Airframe integration paired with a fiber payout spool

Technical Specifications

Parameter / Specification	Value / Detail
Unit Type	Type A — Airborne Sky Unit
Optical Transmission Rate	1.25 Gbps
Optical Connector	FC (single fiber)
Serial / Telemetry	TTL serial (Tx/Rx) via GH1.25
Connector	GH1.25 (8P) back panel
Power Budget	5–12 V DC (via GH1.25)

PASSIVE COMPONENTS

Fiber Optic Adapter

Precision bulkhead adapters that join like connectors for low-loss optical continuity.

Precision passive coupler that joins two like fiber optic connectors end-to-end across a bulkhead, patch panel or enclosure, using a zirconia-ceramic or phosphor-bronze split sleeve available for all common connector standards in SM and MM.

Key Features

- Couples two like connectors (SC, LC, FC, ST, MU, E2000, MT-RJ) for optical continuity across a panel
- High-precision zirconia-ceramic or phosphor-bronze split alignment sleeve for accurate ferrule alignment
- Low insertion loss ≤ 0.2 dB for minimal signal degradation
- High return loss ≥ 50 dB (UPC) and ≥ 60 dB (APC) for single-mode
- Available in simplex, duplex, and quad configurations
- UPC and APC polish options to match connector end-faces
- Compatible with both single-mode (OS1/OS2) and multimode (OM1–OM4) fiber
- Plug-and-play installation with no special tools; ≥ 1000 mating cycles over −40 °C to +75 °C

Applications

- Fiber-to-the-Home (FTTH) and Fiber-to-the-Building (FTTB) network deployments
- Telecommunications patch panels, ODFs, and bulkhead interconnects
- Data centre cross-connects and enterprise LAN cabling
- Cable TV (CATV) and broadband infrastructure

Technical Specifications

Parameter / Specification	Value / Detail
Connector Types	SC, LC, FC, ST, MU, E2000, MT-RJ
Configuration	Simplex / Duplex / Quad
Fiber Type	Single-mode (OS1/OS2) / Multimode (OM1–OM4)
Polish	UPC / APC
Durability	≥ 1000 mating cycles

TEST AND MEASUREMENT EQUIPMENT > VIDEOSCOPE

WiFi Wireless Fiber Endface Microscope

Wireless endface microscope with automatic pass/fail analysis on Windows, Android, and iOS.

Wireless/USB-C fiber endface inspector with automatic pass/fail analysis

Key Features

- WiFi and USB Type-C data transfer with real-time video and automatic pass/fail analysis
- Compatible with Windows, Android, and iOS
- 0.42 µm resolution, 5 MP CMOS sensor, 10X optical magnification, <1 µm particle detection
- More than 50 adapter types including MPO/MTP; custom adapters available
- Built-in rechargeable Li-ion battery, up to 5 hours; usable while charging
- White LED lighting for low-light and dark environments

Applications

- 5G optical network construction and maintenance
- Data centre connector inspection
- High-reliability fiber connections
- Laboratory and manufacturing test
- Endface inspection before OTDR testing

Technical Specifications

Parameter / Specification	Value / Detail
Resolution	0.42 µm
Image Sensor	5-megapixel CMOS
Optical Magnification	10X
Field of View	512 µm × 384 µm
Signal Output	WiFi / USB Type-C
Battery Life	5 hours (rechargeable Li-ion)